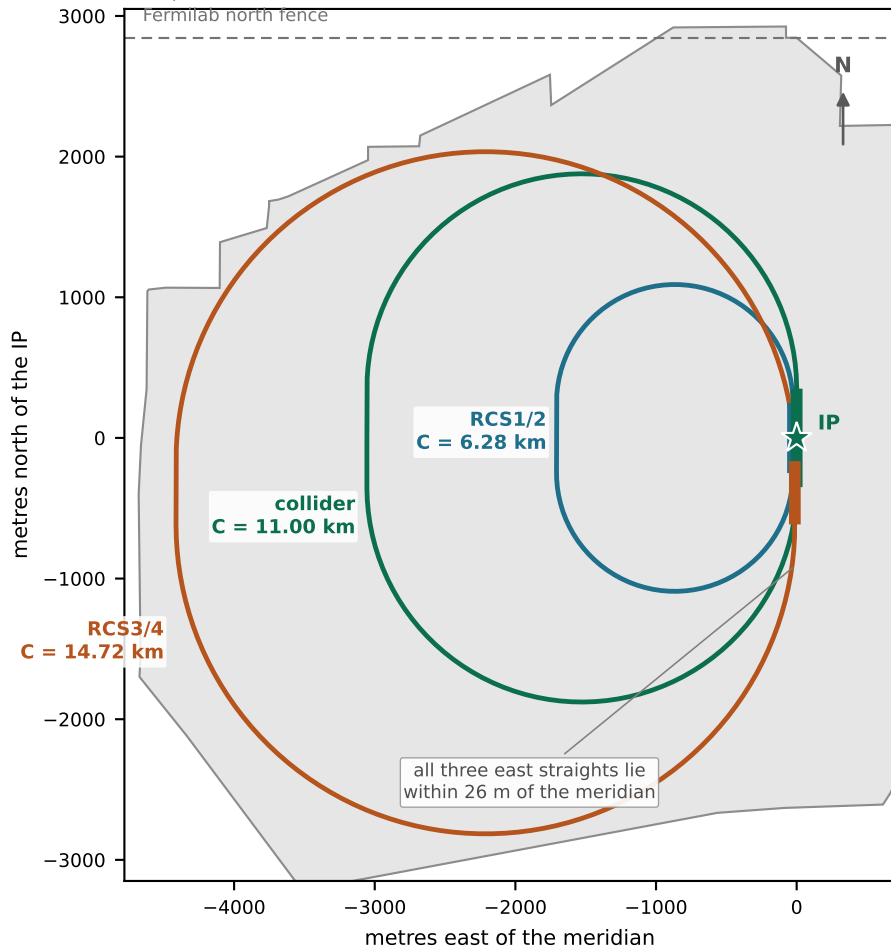


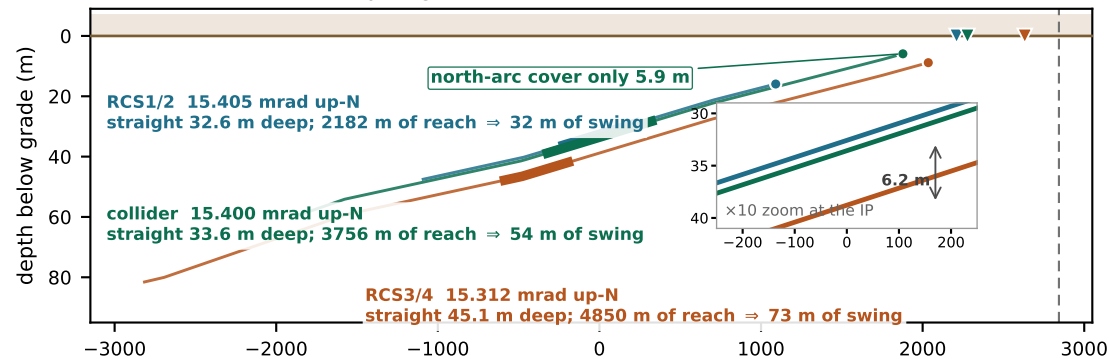
Depth and tilt of the three rings

each ring is a planar racetrack, so its tilt turns the whole north–south reach into depth: $z(y) = z_0 + \theta(y - y_c) + (y - y_c)^2/2R_E$

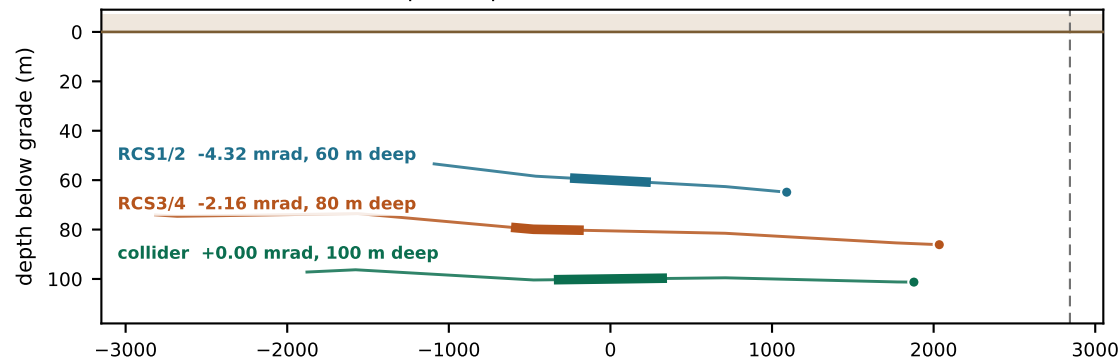
(a) plan: three nested racetracks



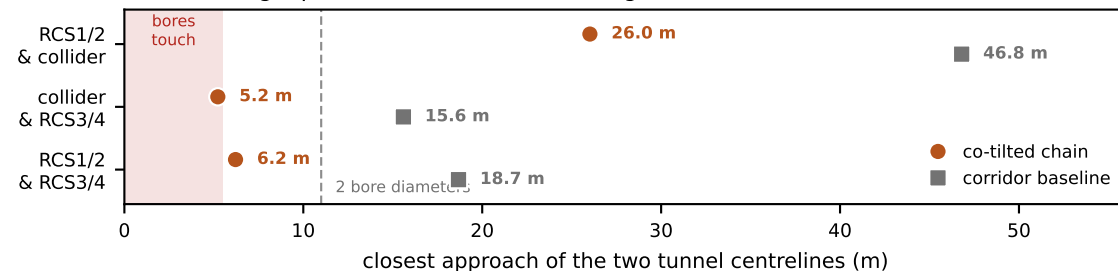
(b) co-tilted chain: every ring aimed at UIUC



(c) corridor baseline: RCS planes pitched down-to-north onto the 9.25 km hall



(d) co-tilting squeezes the three tunnels together



— east straight (on the meridian, aimed at UIUC)
— the rest of the ring, projected onto the section

● north-arc apex: the shallowest cover
▼ where the north pencil reaches the surface
— grade
--- Fermilab north fence

Deepest points 82 m (RCS3/4's south arc), 60 m (collider), 48 m (RCS1/2): the 200 m ceiling never binds a ring. What binds is the shallow end – 5.9 m of cover over the collider's north arc – and, once every ring is co-tilted to the same 15.4 mrad, the vertical room between them: the three planes end up inside a 6.2 m band, so where two rings cross in plan the tunnels pass within 5.2 m. The corridor baseline, whose planes are 20–40 m apart, keeps 16–47 m.